

Meta-analysis

Master of statistics
KU Leuven

Investigating and modelling heterogeneity

Prof. Wim Van den Noortgate
Faculty of Psychology and Educational
Sciences

Leuven - University of Leuven, Belgium
Wim.VandenNoortgate@kuleuven.be



Steps in a meta-analysis

1. Formulating research questions
2. Carefully plan your study
3. Systematic search for studies
4. Selection of relevant studies
5. Extracting relevant data (outcomes, study characteristics, study quality)
6. Converting data to a common effect size measure
7. Combining and comparing the effect sizes
8. Looking for moderator variables
9. Discussing and reporting



Combining & comparing effect sizes

**a. combining effect sizes: the
FEM**

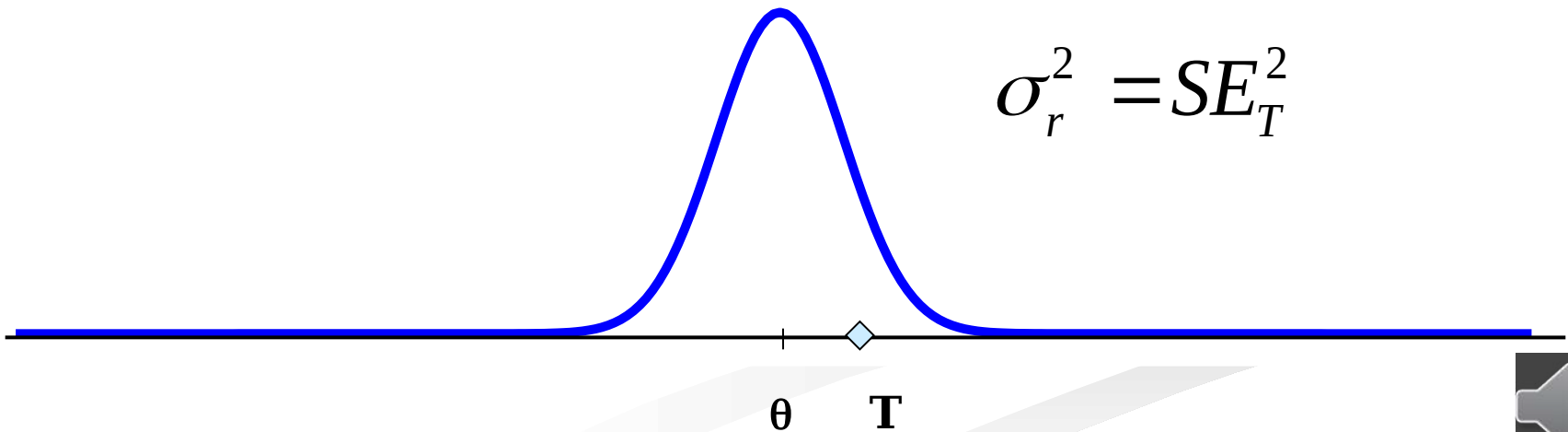


T : observed effect size

θ : 'true' effect size

$$T = \theta + r$$

$$\sigma_r^2 = SE_T^2$$



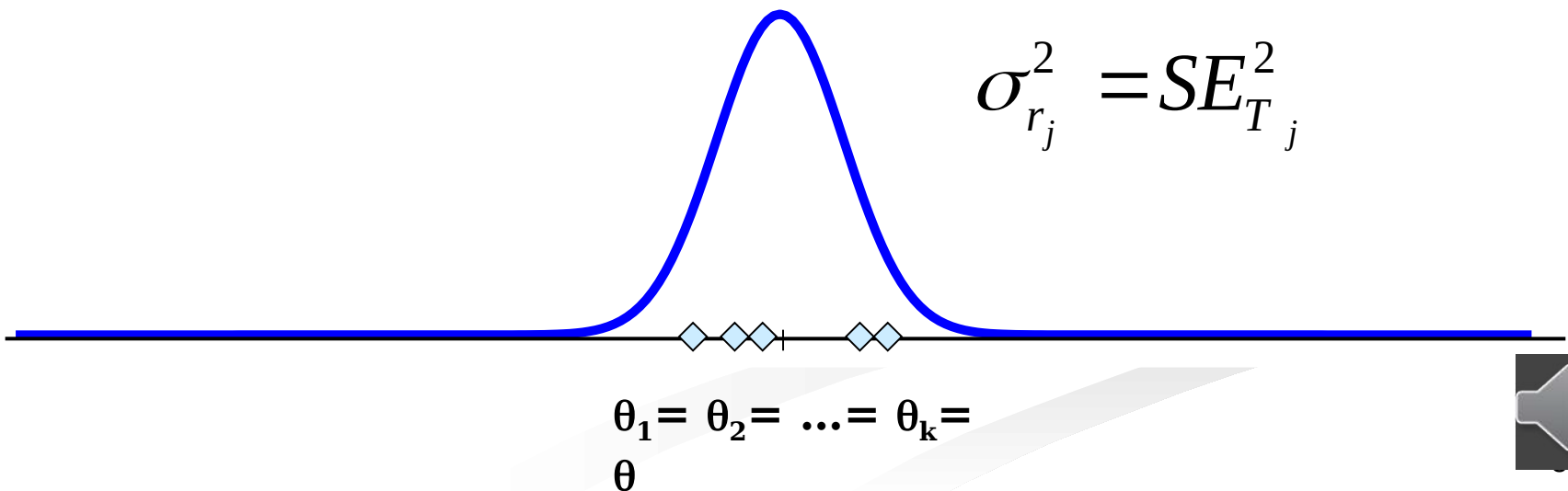
The fixed effects model

Fixed effects model:

$$T_j = \theta_j + r_j$$

$$\theta_1 = \theta_2 = \dots = \theta$$

$$\sigma_{r_j}^2 = SE_{T_j}^2$$



Point estimation of θ

- averaging T's by a simple mean:

$$\hat{\theta} = \frac{\sum_{j=1}^k T_j}{k}$$

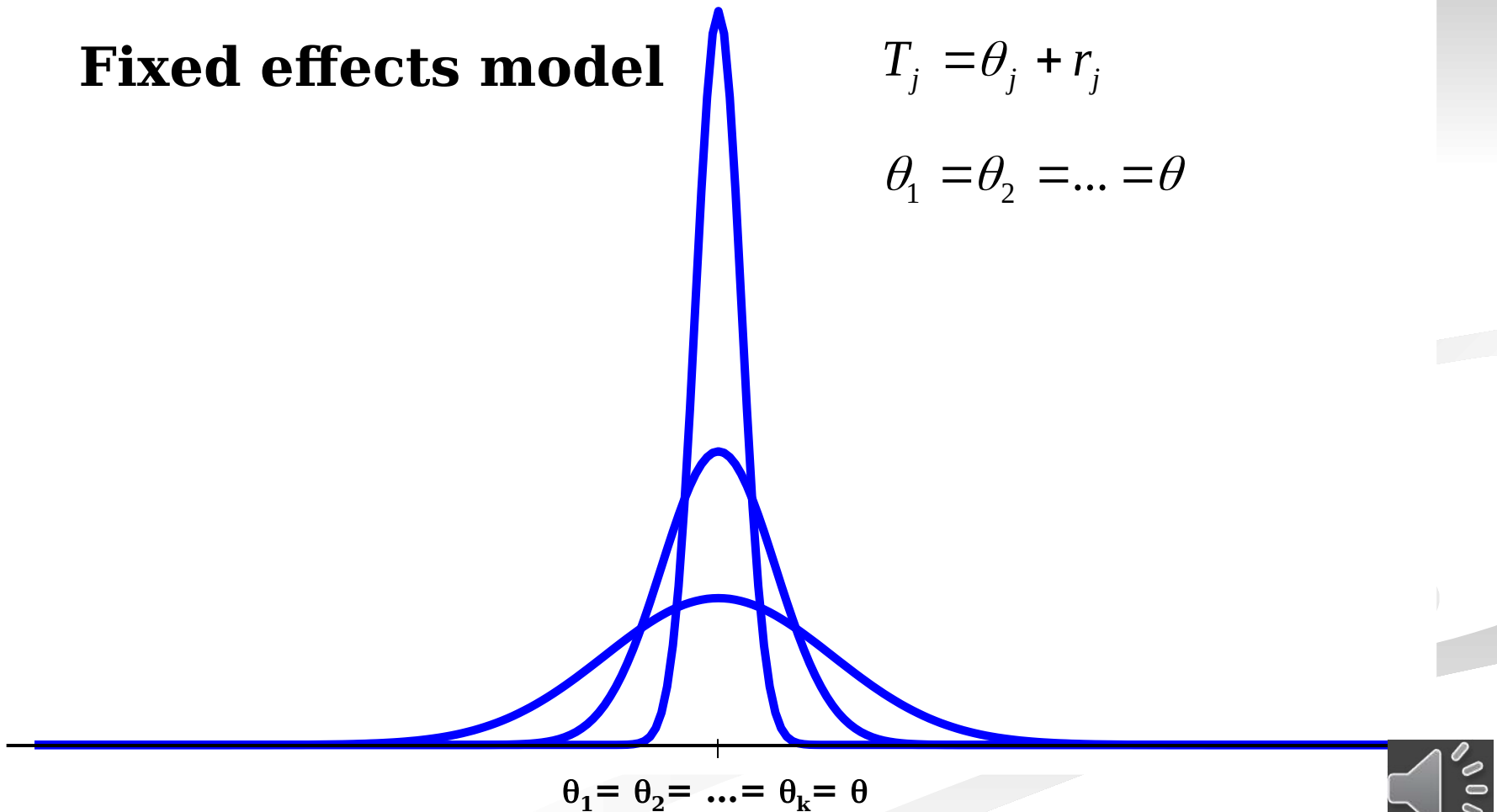
$$\sigma_{\hat{\theta}}^2 = \frac{\sigma_{T_j}^2}{k}$$



Fixed effects model

$$T_j = \theta_j + r_j$$

$$\theta_1 = \theta_2 = \dots = \theta$$



Point estimation of θ

- averaging T's by a simple mean
- or by a precision-weighted mean:

$$\hat{\theta} = \frac{\sum_{j=1}^k w_j T_j}{\sum_{j=1}^k w_j} \quad \text{with } w_j = \frac{1}{\sigma_{T_j}^2}$$

$$\sigma_{\hat{\theta}}^2 = \left(\sum_{j=1}^k \frac{1}{\sigma_{T_j}^2} \right)^{-1}$$

- Sampling variance smallest when using precision as weights



Testing and interval estimation of θ

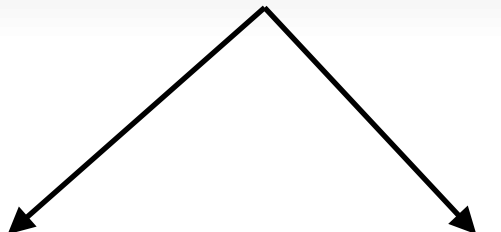
$$T_j \sim N(\theta, \sigma_{T_j}^2)$$

$$\Downarrow$$
$$\hat{\theta} \sim N(\theta, \sigma_{\hat{\theta}}^2) \quad \text{with} \quad \sigma_{\hat{\theta}}^2 = \left(\sum_{j=1}^k \frac{1}{\sigma_{T_j}^2} \right)^{-1}$$

$$\Downarrow$$
$$\frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim N(0,1)$$



Testing and interval estimation of θ

$$\frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim N(0,1)$$


Testing $H_0: \theta = \theta_0$:

$$\frac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}} \sim N(0,1)$$

95% CI:

$$\left[\hat{\theta} - 1.96 * \sigma_{\hat{\theta}}; \hat{\theta} + 1.96 * \sigma_{\hat{\theta}} \right]$$



Combining & comparing effect sizes

**a. combining effect sizes: the
FEM**

b. assessing heterogeneity



... studies often differ in:

- Kind of participants (e.g., age)
- Study context
- Outcome variable and its operationalization
- Independent variable and its operationalization
- Study design
- ...



Many Labs-projects: published studies replicated in multiple laboratories around the world. For 8 out of 16 effects (Klein et al., 2014) and 13 out of 28 effects (Many Labs 2), heterogeneity was found (=variation is larger than you could expect based on sampling variance alone)!

Klein, R. A., Ratliff, K. A., Vianello, M., Adams, R. B., Jr., Bahník, Š., Bernstein, M. J., . . . Nosek, B. A. (2014). Investigating variation in replicability: A “many labs” replication project. *Social Psychology*, 45(3), 142-152.

Many Labs 2: <http://projectimplicit.net/nosek/ML2protocol.pdf>



Homogeneity?

$$H_0 : \theta_1 = \theta_2 = \dots = \theta_k$$

H_a : at least 1 θ_j differs from an other

$$Q = \sum_{j=1}^k w_j (T_j - \hat{\theta})^2$$

$$H_0 : Q \sim \chi^2_{(k-1)}$$



Combining & comparing effect sizes

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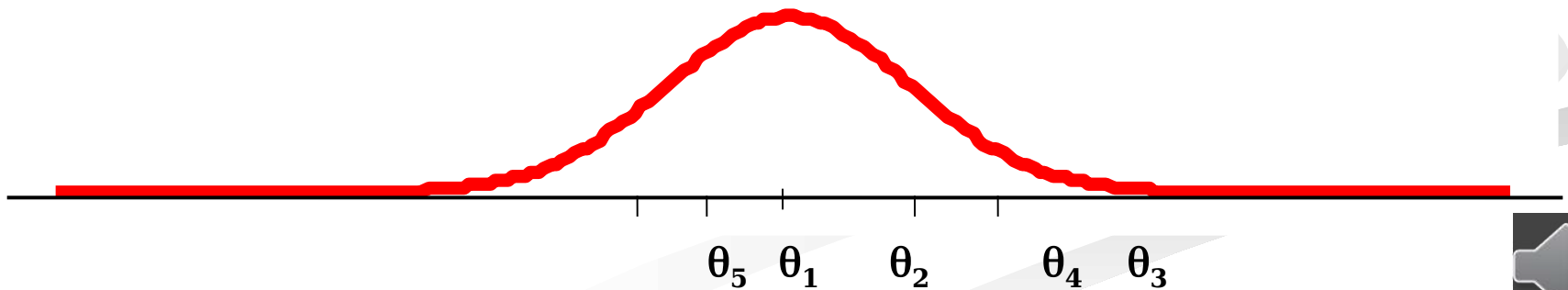
**c. fixed effects vs. random
effects models**



Random effects model

$$T_j = \theta_j + r_j$$

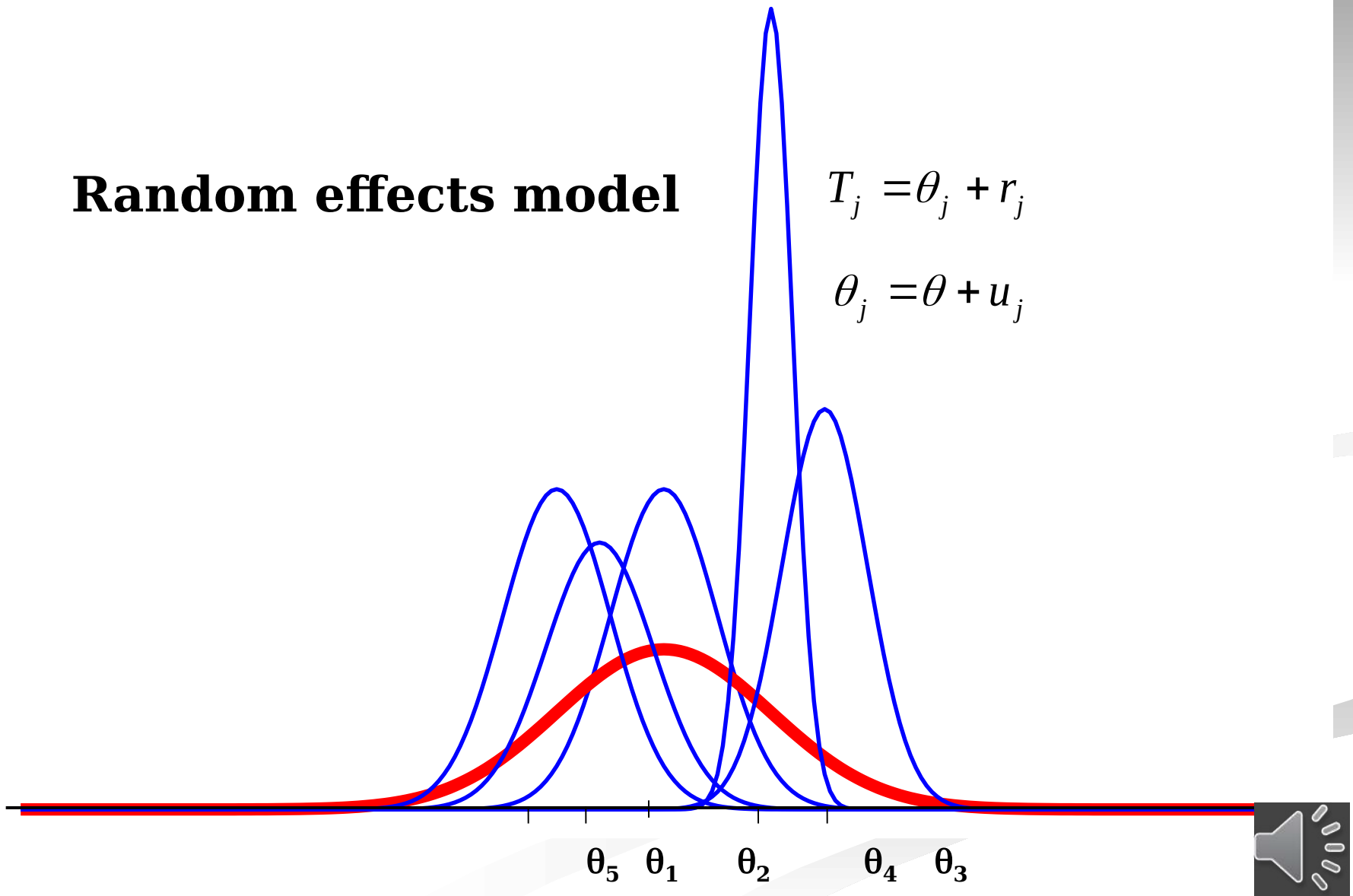
$$\theta_j = \theta + u_j$$



Random effects model

$$T_j = \theta_j + r_j$$

$$\theta_j = \theta + u_j$$



Comparing FEM & REM

The model:

- $T_j = \theta + r_j$
- $T_j = \theta_j + r_j$ and $\theta_j = \theta + u_j$

or $T_j = \theta + u_j + r_j$

What does it mean in words?

Interpretation of θ for a FEM & a
REM?

Parameters to be estimated?



"Of course it mixes apples and oranges; in the study of fruit nothing else is sensible;

comparing apples and oranges is the only endeavor worthy of true scientists;

comparing apples to apples is trivial."

Gene V. Glass, 2000



Value of combining heterogeneous studies?

Contra: mixing apples and oranges

- Heterogeneity as noise
- Hampers the interpretation

Pro

- Allows checking generalization
- Allows nuanced conclusions
- Increases size of meta-analytic data set
($\Rightarrow$ accuracy)



NEW CUYAMA

Population	562
Ft above sea level	2150
Established	1951

TOTAL	4663
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Comparing FEM & REM

Point estimation of θ :

$$\hat{\theta} = \frac{\sum_{j=1}^k w_j T_j}{\sum_{j=1}^k w_j} \quad \text{with } w_j = \frac{1}{\sigma_{T_j}^2}$$

But:

- FEM: $\sigma_{T_j}^2 = SE_{T_j}^2 = \sigma_r^2$
- REM: $\sigma_{T_j}^2 = (SE_{T_j}^2 + \tau^2) = \sigma_r^2 + \sigma_u^2$



In REM:

Consequences for mean effect?

Does this agree with your intuition?



Comparing FEM & REM

Precision of point estimate:

$$\sigma^2_{\hat{\theta}} = \left(\sum_{j=1}^k \frac{1}{\sigma_{T_j}^2} \right)^{-1}$$

But:

- FEM: $\sigma_{T_j}^2 = SE_{T_j}^2 = \sigma_r^2$

- REM: $\sigma_{T_j}^2 = (SE_{T_j}^2 + \tau^2) = \sigma_r^2 + \sigma_u^2$



In REM:

Precision of each observed ES as an estimate of the overall population ES is smaller

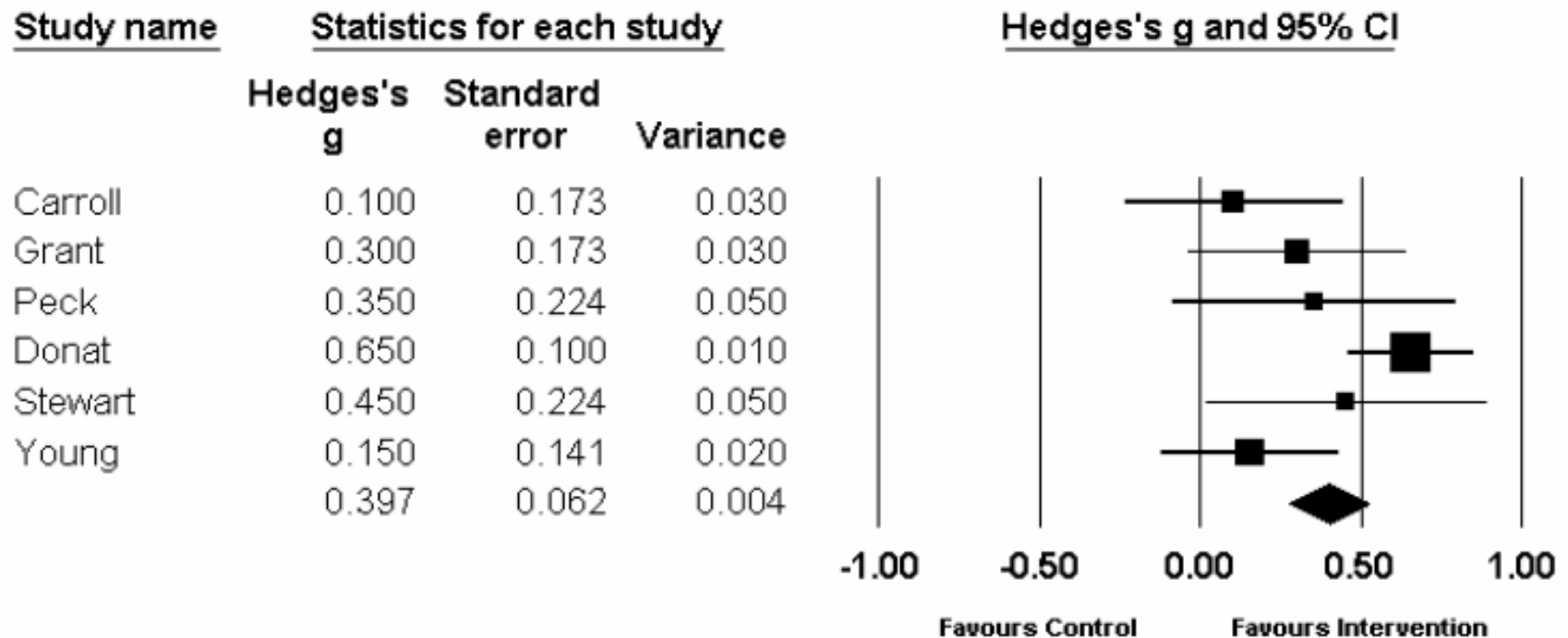
Therefore: precision about the overall estimate is also smaller

Does this correspond with your intuition?



An example

Impact of Intervention - Fixed effect



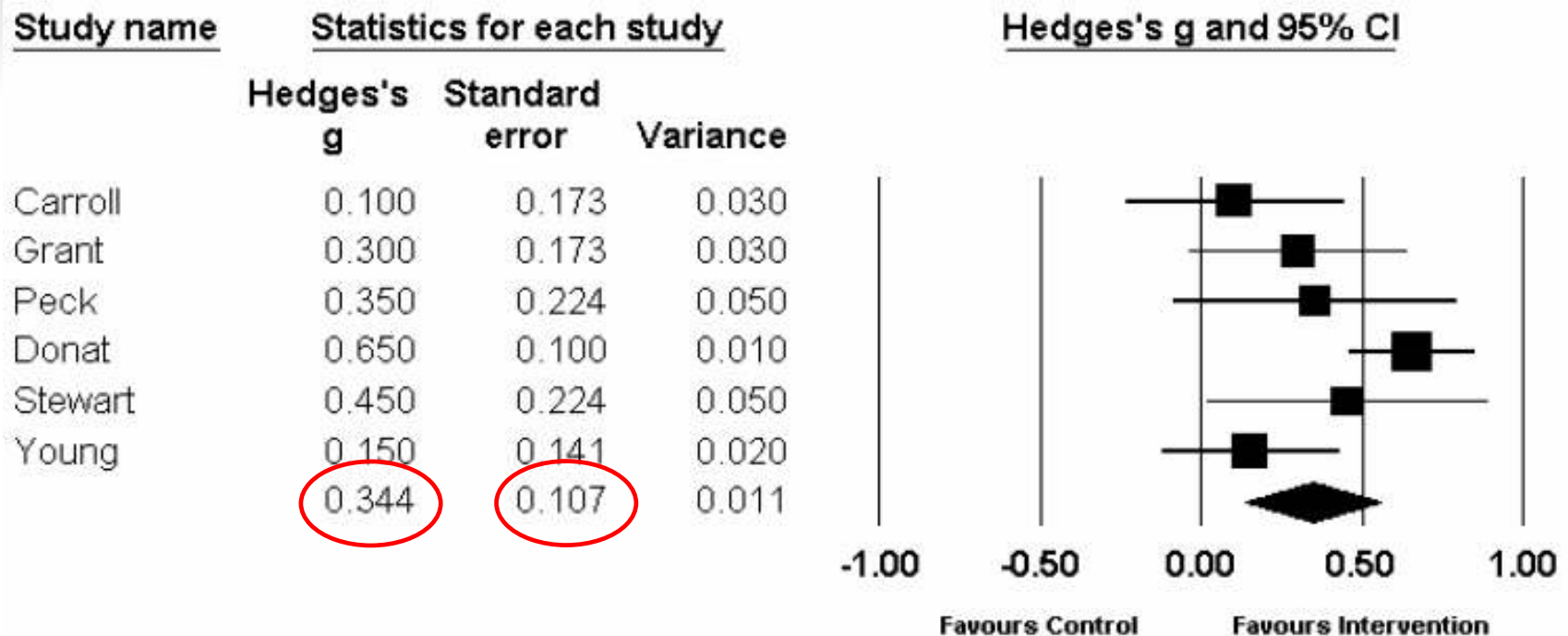
Meta Analysis

Borenstein, M., Hedges, L. V., & Rothstein, H. (2007). *Meta-analysis. Fixed effect vs. random effects*. Retrieved October 10 2007 from www.meta-analysis.com.



An example

Impact of Intervention - Random effects

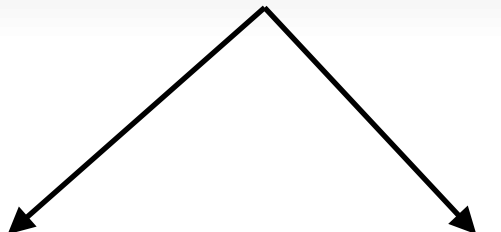


Meta Analysis

Borenstein, M., Hedges, L. V., & Rothstein, H. (2007). *Meta-analysis. Fixed effect vs. random effects*. Retrieved October 10 2007 from www.meta-analysis.com.



Testing and interval estimation of θ

$$\frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim N(0,1)$$


Testing $H_0: \theta = \theta_0$:

$$\frac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}} \sim N(0,1)$$

95% CI:

$$\left[\hat{\theta} - 1.96 * \sigma_{\hat{\theta}}; \hat{\theta} + 1.96 * \sigma_{\hat{\theta}} \right]$$



In REM:

- Lower power
- Smaller accuracy (larger CI)

Then why using the REM?



Estimating τ^2

$$T_j = \theta + u_j + r_j$$

$$\sigma_{T_j}^2 = (SE_{T_j}^2 + \tau^2) = \sigma_r^2 + \sigma_u^2$$

How to disentangle both sources of variance?

$$\hat{\tau}^2 = s_{T_j}^2 - \frac{1}{k} \sum_{j=1}^K \widehat{SE}_{T_j}^2 \quad (\text{Hedges \& Olkin, 1985})$$



The conditional REM approach

- Starts with a FEM
- Performs a homogeneity test
- If heterogeneity, continues with a REM



Discussion:

- Is it reasonable to assume the sample is a *random* sample from a population of effect sizes?
- Do we want to generalize to a broader population?
- Is it reasonable to assume a normal distribution for that population?
- How can we check the underlying distribution?
- What weights are preferable in case of publication bias?



Combining & comparing effect sizes

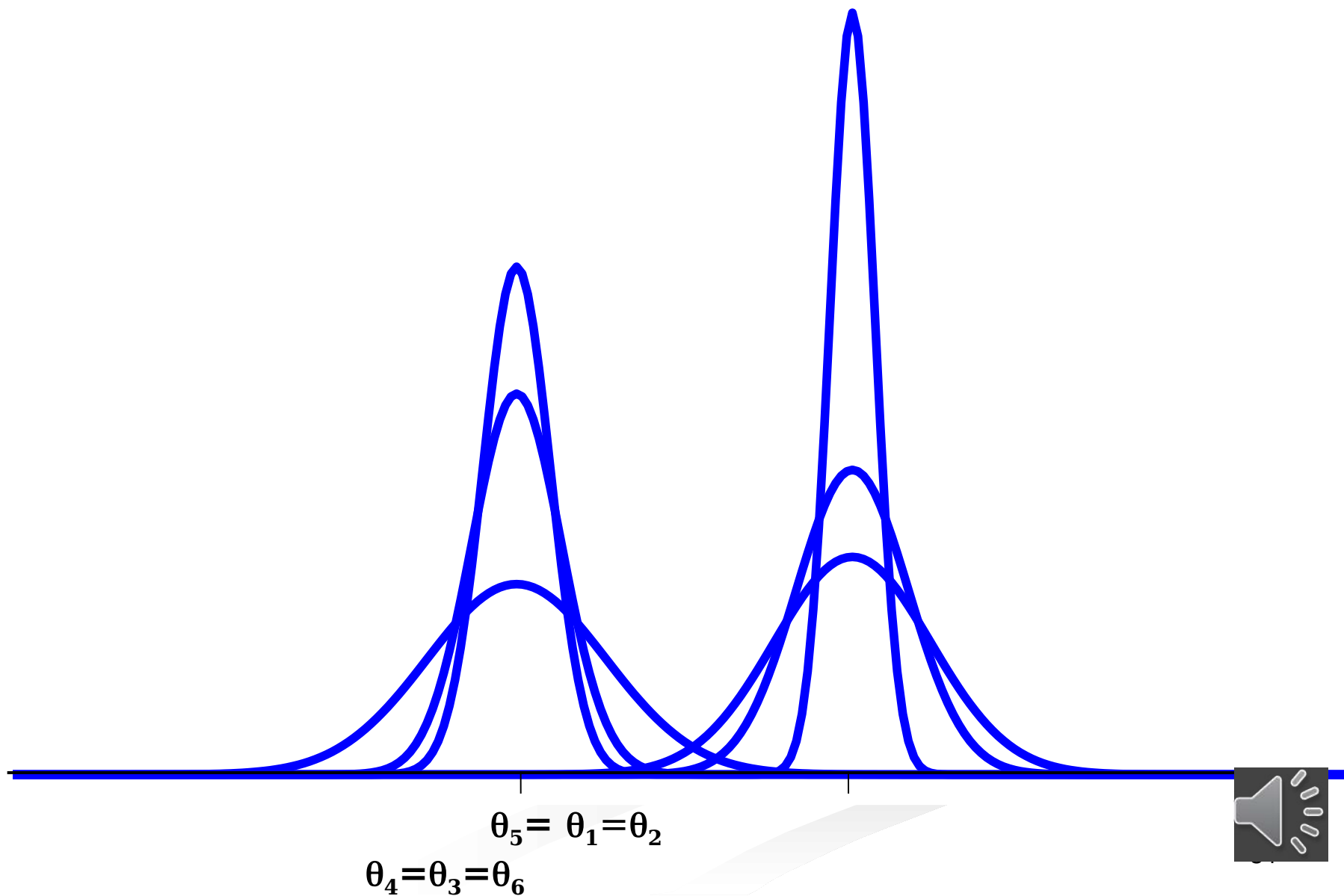
**a. combining effect sizes: the
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**c. fixed effects vs. random effects
models**

**d. understanding differences in
effects**





A weighted ANOVA

$$T_{ij} = \theta_i + r_{ij} \qquad Q = \sum_j \sum_i w_{ij} (T_{ij} - \hat{\theta})^2$$

Total variability
in observed
ES's

=

Variability
between
groups

+

Variability
within
groups

$$Q_T = Q_B + Q_W$$

Q_T : homogeneity test

Q_B : moderator test

Q_W : test for within group
homogeneity

$$H_0: Q_T \sim \chi^2_{k-1}$$

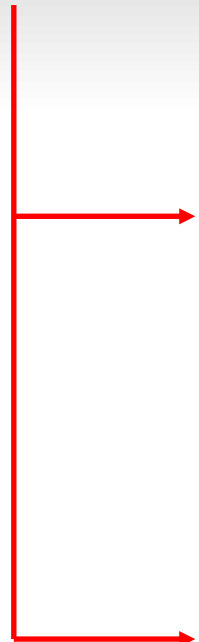
$$H_0: Q_B \sim \chi^2_{I-1}$$

$$H_0: Q_W \sim \chi^2_{k-I}$$



A weighted ANOVA

$$Q_{total} = \sum_j \sum_i w_{ij} (T_{ij} - \hat{\theta})^2$$


$$Q_{between} = \sum_{i=1}^I w_{.i} (\hat{\theta}_i - \hat{\theta})^2$$

$$w_{.i} = \sum_j w_{ij}$$

$$Q_{within} = Q_{within\ 1} + Q_{within\ 2} + \dots + Q_{within\ I}$$

$$Q_{within\ i} = \sum_j w_{ij} (T_{ij} - \hat{\theta}_i)^2$$



What groups differ from each other?

⇒ Testing contrasts

(see e.g., Hedges, L. V. & Pigott, T. D. (2001). The power of statistical tests in meta-analysis. *Psychological Methods*, 6, 203-217.)

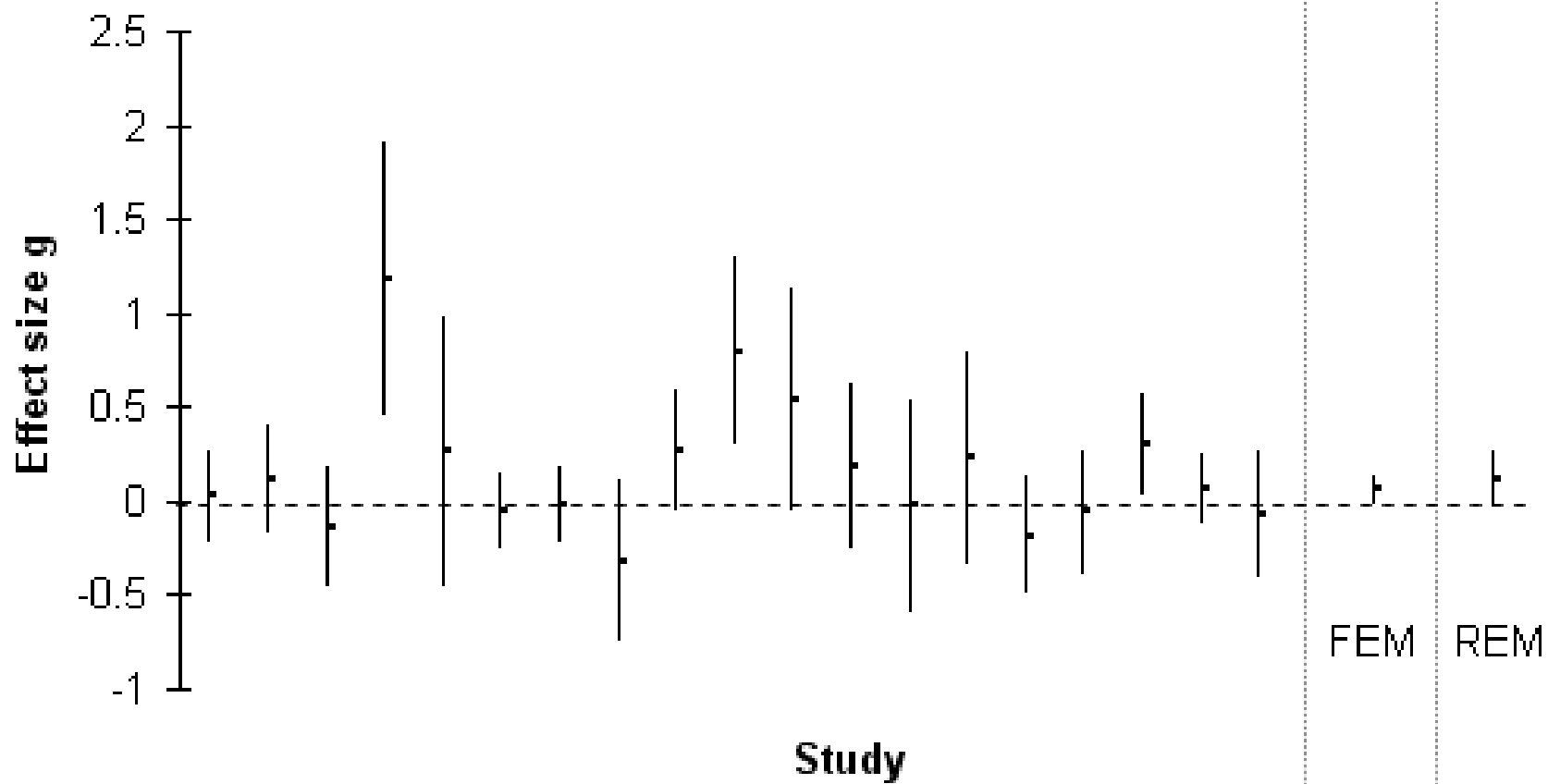


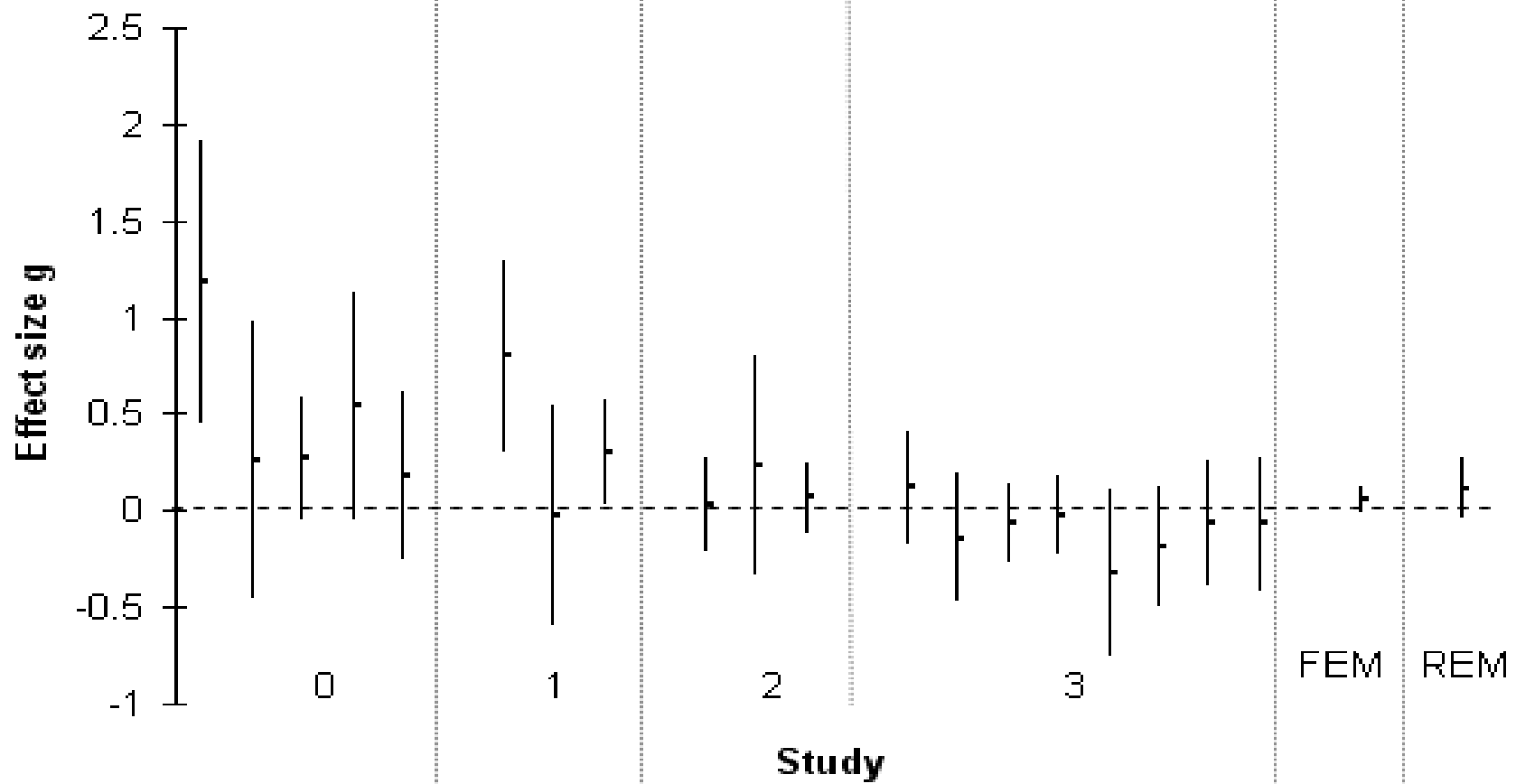
An example in education

(Raudenbush, S. W. (1984). Magnitude of teacher expectancy effects on pupil IQ as a function of the credibility of expectancy induction: A synthesis of findings from 18 experiments. *Journal of Educational Psychology*, 76, 85-97.)

Study	Weeks prior contact	g_j	$\sigma_{(g_j)}$
1. Rosenthal et al. (1974)	2	0.03	0.13
2. Conn et al. (1968)	3	0.12	0.15
3. Jose & Cody (1971)	3	-0.14	0.17
4. Pellegrini & Hicks (1972)	0	1.18	0.37
5. Pellegrini & Hicks (1972)	0	0.26	0.37
6. Evans & Rosenthal (1969)	3	-0.06	0.10
7. Fielder et al. (1971)	3	-0.02	0.10
8. Claiborn (1969)	3	-0.32	0.22
9. Kester & Letchworth (1972)	0	0.27	0.16
10. Maxwell (1970)	1	0.80	0.25
11. Carter (1970)	0	0.54	0.30
12. Flowers (1966)	0	0.18	0.22
13. Keshock (1970)	1	-0.02	0.29
14. Henrickson (1970)	2	0.23	0.29
15. Fine (1972)	3	-0.18	0.16
16. Greiger (1970)	3	-0.06	0.17
17. Rosenthal & Jacobson	1	0.30	0.14
18. (1968)	2	0.07	0.09
19. Fleming & Anttonen (1971)	3	-0.07	0.17







- Cfr. : [‘Raudenbush Example.xls’](#)

	Q total =	Q Between +	Q within
χ^2	35.82537	20.37878	15.44658
df	18	3	15
p	0.007428	0.000142	0.419753

Remark: Hartung's test controls better for Type I error rate



A weighted regression

$$T_j = \theta_j + r_j$$

$$\theta_j = \beta_0 + \beta_1 x_{1j} + \dots + \beta_p x_{pj}$$

= a fixed effects regression model

⇒ a regression analysis, with weights $w_j = \frac{1}{O_{T_j}}$

Under $H_0 : \frac{\hat{\beta}_p}{SE_{\hat{\beta}_p}} \sim N(0,1)$



Combining & comparing effect sizes

**a. combining effect sizes: the
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effects**

e. mixed effects models



- Random effects model with categorical moderator variable:

$$T_{ij} = \theta_{ij} + r_{ij}$$

$$\theta_{ij} = \theta_i + u_{ij}$$

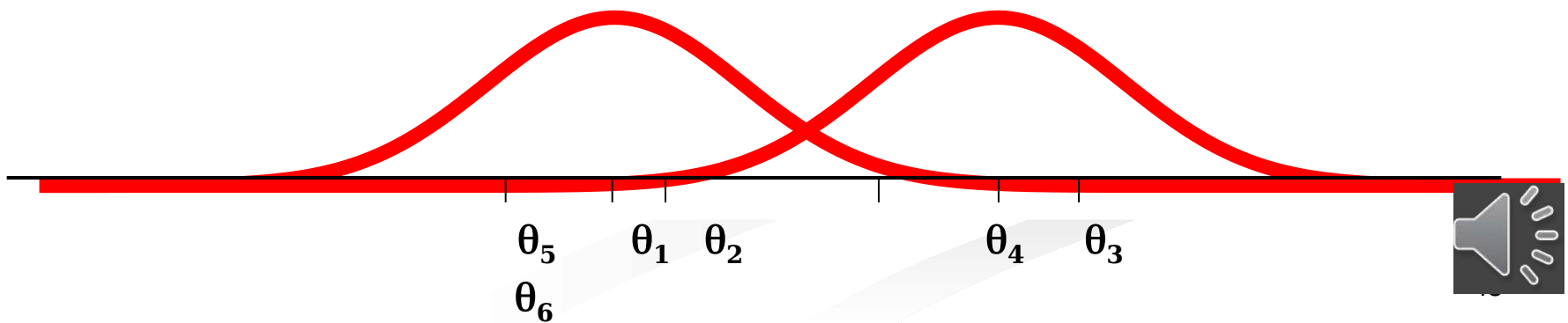
- Random effects model with continuous moderator variable:

$$T_j = \theta_j + r_j$$

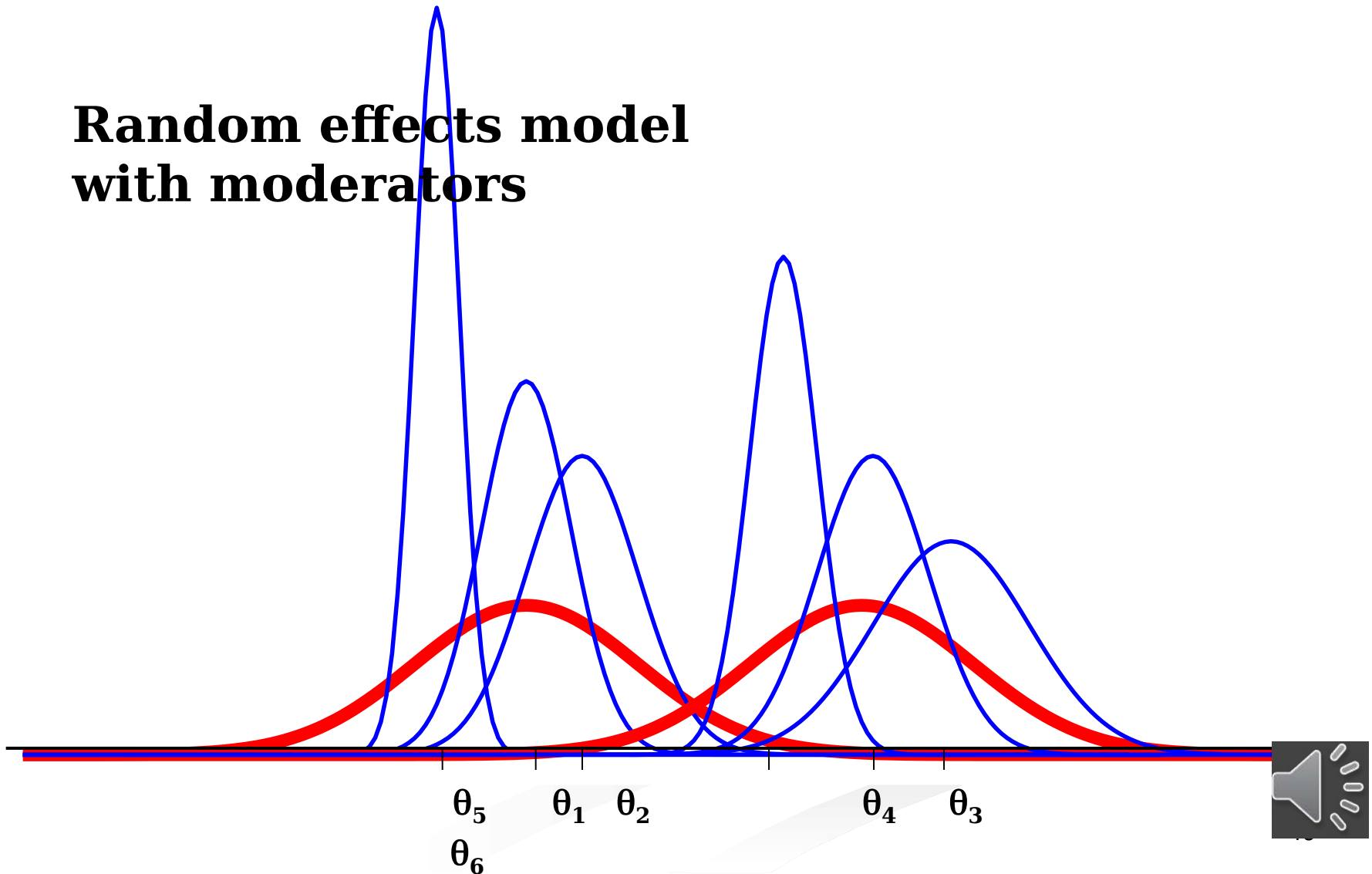
$$\theta_j = \beta_0 + \beta_1 x_{1j} + \dots + \beta_p x_{pj} + u_j$$



Random effects model with moderators



Random effects model with moderators



Random effects model with moderators:

- Random effects regression model or Mixed effects model
- The least restrictive model: allows for moderator variables & random variation



- Remark: typically assumed that within categories the between-study variance is the same (e.g., *metafor*).
- More complex model: different variance per category (e.g., *RevMan*, possibly *CMA*)
- Simulation results: if $\tau_1^2 = \tau_2^2 = \dots = \tau_I^2$, using pooled estimate is better; if not, separate estimate is better

Rubio-Aparicio, M., Sánchez-Meca, J., López-López, J.A., Marín-Martínez, F., & Botella, J. (2017). Analysis of categorical moderators in mixed-effects meta-analysis: Consequences of using pooled vs. separate estimates of the residual between-studies variance. *British Journal of Mathematical and Statistical Psychology*, 70, 439-456.



	FEM	REM
Without moderator		
Categorical moderator		
Continuous moderator		

